通原第5章作业1

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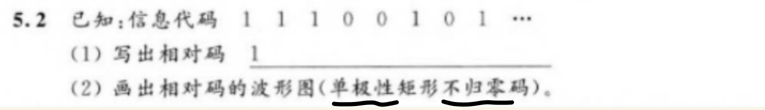


5.2 已知：信息代码 1 1 1 0 0 1 0 1 …

(1) 写出相对码 1________________________

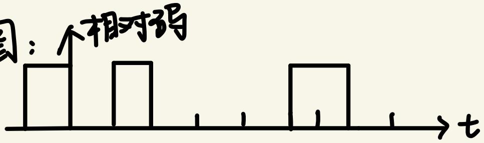
(2) 画出相对码的波形图（单极性矩形不归零码）。

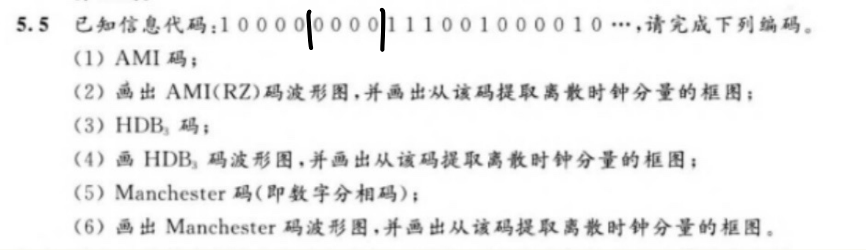
5.2 解：

(1) 1 1 1 0 0 1 0 1

相对码 1 0 1 0 0 0 1 1 0

(2) 波形图：





5.5 已知信息代码：1 0 0 0 0 0 0 0 0 1 1 1 0 0 1 0 0 0 0 1 0 …，请完成下列编码。

(1) AMI 码；

(2) 画出 AMI(RZ)码波形图，并画出从该码提取离散时钟分量的框图；

(3) HDB_3 码；

(4) 画 HDB_3 码波形图，并画出从该码提取离散时钟分量的框图；

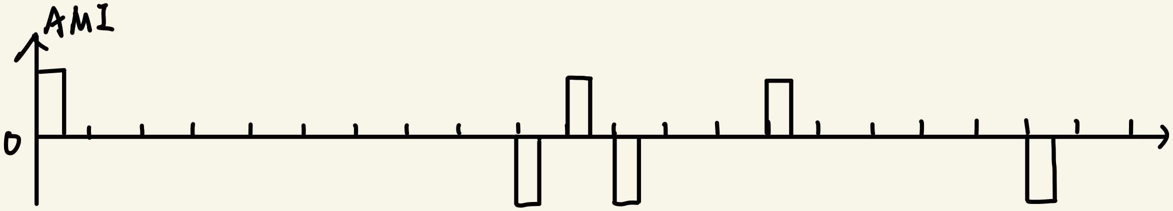
(5) Manchester 码（即数字分相码）；

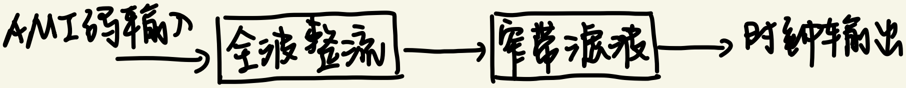
(6) 画出 Manchester 码波形图，并画出从该码提取离散时钟分量的框图。

5.5 解：

(1) AMI码：+1 0 0 0 0 0 0 0 0 -1 +1 -1 0 0 +1 0 0 0 0 -1 0

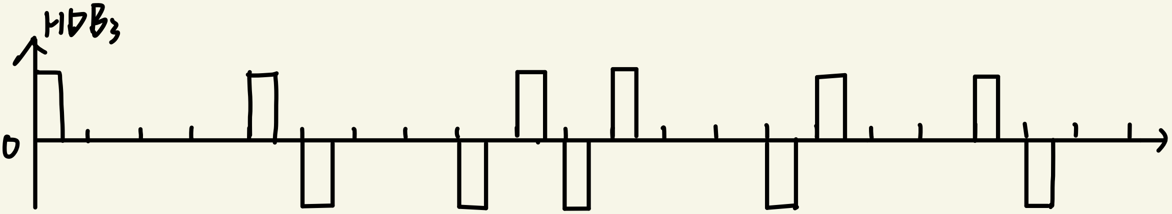
(2) AMI(RZ)码波形：

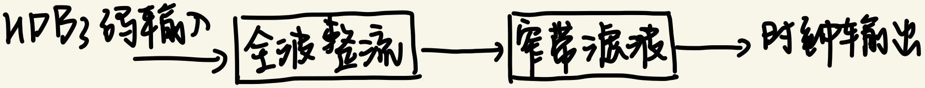




(3) HDB_3码：+1 0 0 0 +V -B 0 0 -V +1 -1 +1 0 0 -1 +B 0 0 +V -1 0

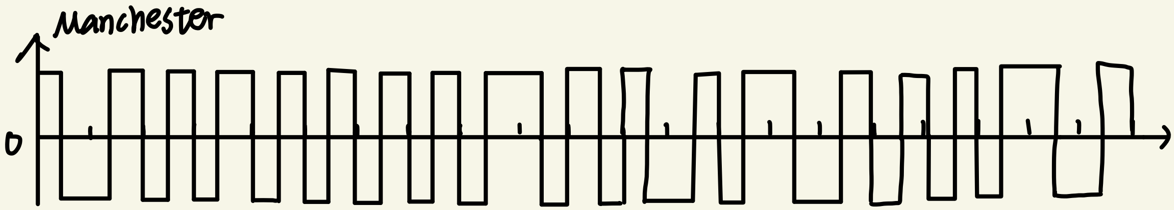
(4) HDB3码波形:

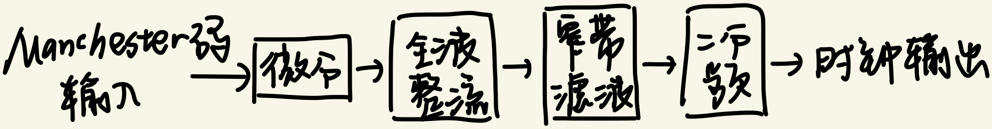


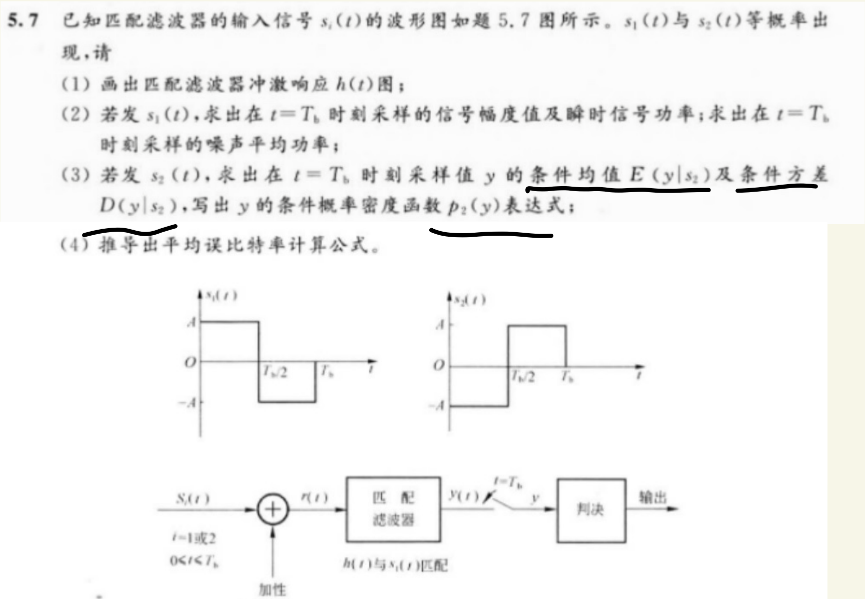


(5) Manchester码: 10 01 01 01 01 01 01 01 01 10 10 10 01 01 10 01 01 01 01 10 01

(6) Manchester码波形:







5.7 已知匹配滤波器的输入信号 $s_i(t)$ 的波形图如题 5.7 图所示。$s_1(t)$ 与 $s_2(t)$ 等概率出现，请

(1) 画出匹配滤波器冲激响应 $h(t)$ 图；

(2) 若发 $s_1(t)$，求出在 $t=T_b$ 时刻采样的信号幅度值及瞬时信号功率；求出在 $t=T_b$ 时刻采样的噪声平均功率；

(3) 若发 $s_2(t)$，求出在 $t=T_b$ 时刻采样值 y 的条件均值 $E(y|s_2)$ 及条件方差 $D(y|s_2)$，写出 y 的条件概率密度函数 $p_2(y)$ 表达式；

(4) 推导出平均误比特率计算公式。

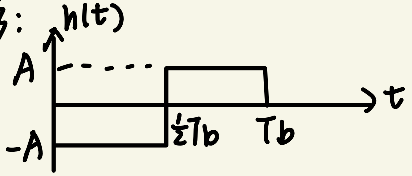
$s_1(t)$
A
O
$-A$
$T_b/2$
T_b
t
$s_2(t)$
A
O
$-A$
$T_b/2$
T_b
t
$S_i(t)$
$i=1$或2
$0<t<T_b$
$r(t)$
匹配
滤波器
$h(t)$与$s_i(t)$匹配
$y(t)$
$t=T_b$
y
判决
输出
加性

5.7解:

(1) $h(t)=S_1(T_b-t)$ $(K=1, t_0=T_b)$

$\Rightarrow h(t)=S_2(t)$

波形:



(2) 当发 $S_1(t)$ 时,

有用信号: $\int_{-\infty}^{+\infty} S_1(T_b-\tau)h(\tau)d\tau = \int_{-\infty}^{+\infty} S_2(\tau)S_2(\tau)d\tau = A^2T_b = E_b$

且 $E_b=E_1=E_h$

信号分量功率 $=E_b^2$

噪声分量功率 $\sigma^2=\frac{N_0}{2}\cdot E_h=\frac{N_0}{2}\cdot E_b$

(3) 当发 $S_2(t)$ 时,

有用信号: $\int_{-\infty}^{+\infty} S_2(T_b-\tau)h(\tau)d\tau = -\int_{-\infty}^{+\infty} S_2(\tau)S_2(\tau)d\tau = -A^2T_b=-E_b$

∴判决量 $y=-E_b+z\sim(-E_b,\sigma^2)$, 为高斯随机变量

即条件均值为 $-E_b=-A^2T_b$, 条件方差为噪声方差 $\sigma^2=\frac{N_0E_b}{2}$.

∴ $p_2(y)=p(y|s_2)=\frac{1}{\sqrt{\pi N_0E_b}}e^{\frac{-(y+E_b)^2}{N_0E_b}}$

(4) $p_1(y)=p(y|s_1)=\frac{1}{\sqrt{\pi N_0E_b}}e^{\frac{-(y-E_b)^2}{N_0E_b}}$, 此时 $y=E_b+z\sim(E_b,\sigma^2)$

∵ s_1 和 s_2 等概出现, $P(s_1)=P(s_2)$

$V(T)=\frac{\sigma^2}{2E_b}\ln\frac{P(S_2)}{P(S_1)}$

∴ $V_T=0$

∵平均误比特率 $P_b=P(s_1)P(e|s_1)+P(s_2)P(e|s_2)$

$=P(s_1)\int_{-\infty}^{V_T}p(y|s_1)dy+P(s_2)\int_{V_T}^{\infty}p(y|s_2)dy$

$=\frac{1}{2}P(e|s_1)+\frac{1}{2}P(e|s_2)$

$$p(e|s_1) = p(y<V_T|s_1) = \frac{1}{2}\operatorname{erfc}\left(\frac{E_b - V_T}{\sqrt{2\sigma^2}}\right) = \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$p(e|s_2) = p(y>V_T|s_2) = \frac{1}{2}\operatorname{erfc}\left(\frac{E_b + V_T}{\sqrt{2\sigma^2}}\right) = \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$\therefore P_b = \frac{1}{2}p(e|s_1) + \frac{1}{2}p(e|s_2)$$

$$= \frac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$